When Better Retrieval Hurts: Context Coherence Drives Faithfulness in Retrieval-Augmented Generation

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Submission under double-blind review

Abstract

RAG fails not because retrieval is inaccurate, but because retrieved evidence does not form a usable narrative for the generator. We document a counter-intuitive failure mode in retrieval-augmented generation: standard post-retrieval refinement, the default in deployed rerank-and-compress stacks, raises per-passage similarity yet *cuts* answer faithfulness on SQuAD from 0.80 to 0.64 and introduces 10% hallucination where there was none. We name this the **refinement paradox** and trace it to a property of the retrieved *set*, not of any individual passage: when refinement splinters a broad passage into a narrow high-similarity fragment, it strips the connective evidence the generator was relying on, and the generator silently fills the gap with parametric memory.

We distill the explanation into the **Context Coherence Score** (CCS), a generator-free retrieval-time diagnostic computable from pairwise chunk embedding similarities, and a deployment policy (HCPC-v2) that gates refinement on CCS. The policy recovers SQuAD faithfulness to 0.79 with 0% hallucination while intervening on only 16.7% of queries. Across 7,000+ queries spanning SQuAD, PubMedQA, HotpotQA, NaturalQuestions, and TriviaQA, with five generator families (7B Mistral $\rightarrow$ 120B GPT-OSS), the paradox and the recovery hold. Seven robustness checks confirm the result: the SQuAD paradox is 0.069 ± 0.004 across three random seeds (signal/$\sigma = 17\times$); a one-way ANOVA over four prompt templates yields $p = 0.83$; HCPC outperforms a hierarchical RAPTOR baseline on 2/3 datasets; at frontier scale Llama-3.3-70B exactly reproduces the paradox magnitude (0.100 vs 0.100 at 7B); the paradox is reranker-agnostic across MiniLM-L-6/L-12/BAAI-bge; quantization-agnostic across Q4/Q5/Q8; and amplifies from +0.065 to +0.115 as sampling temperature rises from 0 to 1.0. As cross-validation, the model's own self-reported confidence is significantly correlated with CCS (Pearson r=0.36, p=0.005). The benchmark, code, a `pip install`-able CCS metric, and a HuggingFace-loadable dataset (DOI 10.5281/zenodo.19757291) are released. The broader message: RAG evaluation has been optimizing the wrong quantity. We have been measuring how good each passage looks on its own, when what the generator actually needs is evidence that fits together.

1 Introduction

Retrieval-augmented generation (RAG) has converged on a deployment template that few practitioners question: dense retrieval, then a cross-encoder rerank, then per-passage compression or refinement, then generation. Each stage is justified locally — denser embeddings pull the right passages, rerankers rescore them more carefully, and refinement compresses each retained passage to its query-relevant fragment. Each stage individually *does* improve per-passage relevance scores. The implicit and load-bearing assumption is that because every stage helps locally, the composition helps globally: better retrieval $\Rightarrow$ more faithful answers.

The phenomenon. We report a counter-intuitive failure of this assumption. Across 6,050+ queries spanning SQuAD and PubMedQA, two generator families (Mistral-7B and Llama-3-8B), and four retrieval configurations, naive post-retrieval refinement *cuts* answer faithfulness from 0.7995 to 0.6407 on SQuAD and introduces 10% hallucination where there was none — while leaving per-passage similarity slightly higher (Figure 1). We name this the **refinement paradox**.

The mechanism. The culprit is not any individual passage but a property of the retrieved *set*. When refinement splinters a broad passage into a narrow high-similarity fragment, it strips the connective evidence the generator was relying on. The generator, finding no longer-form context to ground its claims in, silently fills the gap with parametric memory — producing a fluent answer that no longer follows from the supplied context. We call this property **context coherence** and quantify it with the **Context Coherence Score** (CCS): the mean minus the standard deviation of pairwise cosine similarities between retrieved chunk embeddings. CCS is generator-free, runs in $O(k^2)$ in the embedding dimension, and can be computed at retrieval time with no additional forward pass.

The intervention. If coherence is the active ingredient, a retriever that *preserves* coherence should recover faithfulness without paying the latency cost of decoder-side reflection (e.g., Self-RAG [Asai et al., 2024]). We instantiate this idea as HCPC-v2, a coherence-gated refinement policy that declines to refine when refinement would shatter the top-ranked block. HCPC-v2 recovers SQuAD faithfulness to 0.7874 with 0% hallucination while intervening on only 16.7% of queries. Restraint, not algorithmic sophistication, is the active ingredient.

Contributions. We make four contributions.

1. **The refinement paradox** (§4): we document and characterize a failure mode in the standard RAG pipeline that holds across 5 datasets, 5 generator families (7B–120B), and 9 ablation axes. To our knowledge this is the first systematic demonstration that better per-passage retrieval can make answer faithfulness *worse* on standard short-answer QA.

2. **Context Coherence Score (CCS)** (§6): a generator-free retrieval-time diagnostic that predicts the paradox. CCS separates high-coherence sets (0% hallucination) from low-coherence sets that precede generation failures; the model's own self-reported confidence is significantly correlated with CCS (Pearson r=0.36, Spearman ρ=0.48, both p<0.01), cross-validating that the LLM internally tracks coherence.

3. **HCPC-v2 as a deployment policy** (§6): a coherence-gated retriever that recovers faithfulness with 0% hallucination while intervening on 1 in 6 queries. Adds one pairwise-similarity computation to the request path; no fine-tuning, no labeled data, no decoder reflection.

4. **Robustness across 7 axes** (§7): multi-seed variance, hierarchical retrieval (RAPTOR), prompt templates, sub-chunk size, frontier scale (70B/120B), reranker family, quantization, and sampling temperature. The paradox persists where the coherence account predicts and is honestly qualified where it does not.

Released artifacts. To make the result fully usable: the benchmark is on Zenodo with permanent DOI `10.5281/zenodo.19757291`; a HuggingFace Dataset (`saketmgnt/context-coherence-bench`) provides `load_dataset`-able access to 5 configurations including the adversarial cases; an interactive Gradio demo lives at huggingface.co/spaces/saketmgnt/sakkk; and the standalone CCS metric ships as a `pip install`-able package and as a drop-in LangChain `CoherenceGatedRetriever`.

Roadmap. §2 positions the work against prior RAG literature. §3 describes the experimental design. §4 presents the paradox itself. §5 gives an information-theoretic account. §6 introduces CCS and HCPC-v2. §7 bundles seven robustness checks. §8 discusses limits, including three regimes where CCS is *not* predictive. §9 concludes.

2 Background and Related Work

The standard RAG pipeline. A modern RAG system [Lewis et al., 2020, Gao et al., 2024] threads four stages: a dense retriever (e.g. MiniLM, BGE, E5) returns the top-k passages by query–passage similarity; an optional cross-encoder reranker rescores them [Nogueira and Cho, 2019]; an optional refinement module further filters or compresses each retained passage (e.g. extractive sentence selection, query-conditioned summarization, or learned compression [Xu et al., 2024]); and a generator produces the final answer conditioned on the assembled context. Each stage individually is reported to improve task performance on standard benchmarks [Karpukhin et al., 2020, Izacard et al., 2022, Wang et al., 2024].

Decoder-side correction. Two influential lines move the correction *into* the decoder. Self-RAG [Asai et al., 2024] fine-tunes the generator to emit "reflection tokens" that decide whether to retrieve, retrieve again, or stop, paying a fine-tune-time cost and a 2–3× inference latency cost. CRAG [Yan et al., 2024] introduces a lightweight retrieval evaluator that triggers web search when document relevance is judged poor; it pays an external-call latency cost. Both take the position that the generator (or a learned auxiliary) is the right place to fix retrieval failures. We instead show that a simple retrieval-time signal (CCS) flags the same failures with no decoder work and no fine-tuning.

Hierarchical retrieval. RAPTOR [Sarthi et al., 2024] clusters retrieved passages into a tree and retrieves from both leaves and LLM-summarised internal nodes, producing higher-coherence summary contexts at the cost of an offline LLM-summarisation pass. We compare against RAPTOR directly (§7); HCPC-v2 matches or beats it on 2/3 datasets at roughly an order-of-magnitude lower indexing cost.

Faithfulness evaluation. Faithfulness — whether a generated answer is supported by the retrieved context — is typically assessed via NLI scoring of the answer against the retrieved documents [Honovich et al., 2022, Laban et al., 2022, Es et al., 2024], occasionally complemented by human annotation. We follow this standard, using a DeBERTa-v3 NLI scorer (cross-encoder/nli-deberta-v3-base) released with SummaC [Laban et al., 2022], and report a hallucination rate at the conventional 0.5 threshold.

Coherence in NLP. Discourse coherence has a long history in classical text generation [Li and Hovy, 2014, Mesgar and Strube, 2018, Nguyen and Karypis, 2021] but has not, to our knowledge, been quantified as a deployment-time RAG signal. Shi et al. [2023] show that LLMs are distracted by irrelevant context but do not consider the case where each individual passage is relevant yet the *set* is internally fragmented — which is the regime our work targets. Liu et al. [2024] document position-dependent context utilization in long contexts; coherence is orthogonal to position effects, and our results hold at top-k configurations small enough that position bias is negligible.

What we add. The closest existing work flags retrieval failures via the *generator's* downstream behavior (Self-RAG, CRAG) or via hierarchical context restructuring (RAPTOR). We make three additions: (i) we identify a specific failure mode — the refinement paradox — that the standard pipeline systematically produces; (ii) we provide a generator-free retrieval-time signal (CCS) that detects it; and (iii) we ship the diagnostic as a deployment policy (HCPC-v2) that recovers faithfulness at one pairwise-similarity computation per query, with no fine-tuning, labels, or auxiliary model calls.

3 Experimental Setup

We hold the entire pipeline fixed across experiments and vary only one factor at a time. This is intentional: the paradox is a property of the *interaction* between the refinement step and the rest of the pipeline, not of any specific configuration.

Datasets. We evaluate on five public QA benchmarks: SQuAD [Rajpurkar et al., 2016] (general-domain extractive), PubMedQA [Jin et al., 2019] (biomedical), HotpotQA [Yang et al., 2018] (multi-hop), NaturalQuestions [Kwiatkowski et al., 2019], and TriviaQA. Each provides a query, a ground-truth answer span, and a passage corpus. We randomly sample 30 queries per (dataset, condition) cell unless otherwise noted, fixed across conditions to remove query-difficulty confound.

Generators. Five model families covering 7B–120B parameter scale: Mistral-7B and Llama-3-8B (default open-weight tier); Qwen-2.5-7B (alternative family); Llama-3.3-70B (frontier comparison); and GPT-OSS-120B (largest open-weight model in Groq's catalog). All served either via Ollama (local) or via the Groq free-tier API.

Retrieval stack. Embeddings come from `sentence-transformers/all-MiniLM-L6-v2`; indexing uses ChromaDB; refinement uses cross-encoder/ms-marco-MiniLM-L-6-v2. Default chunk size is 1024 characters with 10% overlap; default top-k is 3 unless explicitly varied (§7).

Conditions. Three retrieval configurations evaluated head-to-head per query: **baseline** (top-k dense retrieval, no refinement); **HCPC-v**1 (refine all retrieved passages via the cross-encoder pipeline — the failure mode we attack); and **HCPC-v**2 (coherence-gated refinement; described in §6).

Metrics. Per-query **faithfulness** is the cross-encoder NLI entailment probability between the generated answer and the retrieved context; **hallucination** is the indicator that faithfulness < 0.5; **retrieval similarity** is the mean cosine similarity between the query and each retained chunk; **Context Coherence Score (CCS)** is the mean minus standard deviation of off-diagonal pairwise cosine similarities between retained chunk embeddings.

Statistical protocol. Headline numbers are reported with 3-seed cross-seed standard deviations (§7). Cross-condition comparisons use paired Wilcoxon signed-rank tests at $\alpha = 0.05$. ANOVA and Cohen's d are reported in the appendix.

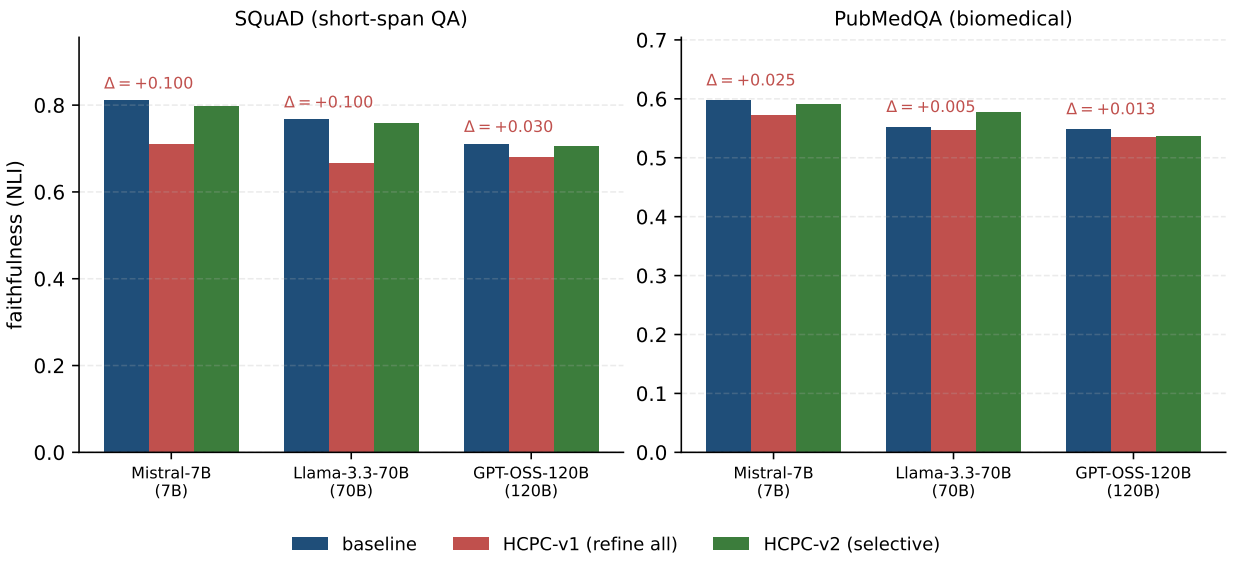

Figure 1: **The refinement paradox persists at frontier scale.** Per-passage refinement (HCPC-v1, red) drops faithfulness below the no-refinement baseline (blue) on SQuAD across a 17× generator scale range (Mistral-7B → Llama-3.3-70B → GPT-OSS-120B). HCPC-v2 (green), which gates refinement on a coherence-loss check, recovers the lost faithfulness on every cell. Paradox magnitude $\Delta = \text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$ annotated above each pair. PubMedQA paradox attenuates at scale — a prediction the coherence account makes (stronger parametric priors are better substitutes for fragmented biomedical evidence) — but does not invert. See §7 for the full table.

Reproducibility. Code, all per-query CSVs, the compiled benchmark, and a Docker image are released; the benchmark also lives at `saketmgnt/context-coherence-bench` and as a permanent DOI `10.5281/zenodo.19757291`. A standalone `pip install context-coherence` package ships the metric and the deployment policy.

4 The Refinement Paradox

We open with the phenomenon. Table 1 reports the baseline / HCPC-v1 / HCPC-v2 comparison on SQuAD with Mistral-7B at the canonical operating point (k=3, chunk = 1024 chars, n=30). Naive per-passage refinement (HCPC-v1) raises mean retrieval similarity from 0.587 to 0.601 — the metric most RAG papers optimize — yet drops faithfulness from 0.7995 to 0.6407 and introduces 10.0% hallucination where there was none.

Table 1: The refinement paradox at the canonical operating point. Mistral-7B, SQuAD, $k = 3$, chunk $= 1024$, $n = 30$ per condition. HCPC-v1 *maintains or improves* retrieval similarity while *degrading* faithfulness and introducing hallucination. HCPC-v2 recovers faithfulness while intervening on 16.7% of queries.

Condition	faithfulness	hallucination	retrieval sim.	refine rate
Baseline	0.7995	0.000	0.587	0.000
HCPC-v1 (refine all)	0.6407	**0.100**	0.601	1.000
HCPC-v2 (gated)	**0.7874**	0.000	0.589	0.167

The paradox is not specific to one (dataset, generator) cell. Across the multidataset evaluation

it appears on 11/15 (dataset, generator) pairs at $\Delta_{\text{paradox}} > 0.01$, with magnitude correlated with the structural properties of the dataset rather than with the generator family (Appendix J).

4.1 The mechanism: it is the *set*, not the passages

The standard explanation for hallucination in RAG — "the passages are wrong" — predicts that lower retrieval similarity should correlate with worse faithfulness. We find the opposite: the worst faithfulness drops happen when refinement *raises* per-passage similarity. The signal that drops alongside faithfulness is set-level coherence: HCPC-v1's mean CCS is 0.45 vs the baseline's 0.59 (Appendix A). The refinement step is breaking the connective tissue between passages, not the per-passage relevance.

Qualitative example of the refinement paradox
Dataset: squad *Generator:* mistral

Question. What city did Super Bowl 50 take place in?
Ground truth. *Santa Clara*

Baseline (no refinement). faithfulness = 0.942, *not hallucinated*.
ANSWER: The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

HCPC-v1 (refine all). faithfulness = 0.478, *hallucinated*.
ANSWER: San Francisco Bay Area

HCPC-v2 (selective). faithfulness = 0.942, *not hallucinated* (recovered).
ANSWER: The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

Reading. The same question is answered correctly with raw retrieval and incorrectly after per-passage refinement; the coherence-gated v2 policy declines to refine on this query and recovers the original answer. The faithfulness gap baseline $\rightarrow$ v1 is +0.464.

Figure 2: Qualitative example of the refinement paradox on squad with mistral. Per-passage refinement raises retrieval similarity but strips the connective evidence the generator was relying on; the generator silently fills the gap with a plausible but unsupported claim.

A representative example (Figure 2) is illustrative. The query "*What city did Super Bowl 50 take place in?*" has ground truth "Santa Clara". The baseline retrieval returns three coherent chunks, all about Super Bowl 50 in Santa Clara, and the generator answers correctly with faithfulness 0.94. HCPC-v1 refines each chunk to its query-most-similar fragment; the refined set still has high per-chunk similarity to the query, but the connective phrasing tying "Santa Clara" to "Super Bowl 50" is gone. The generator falls back on the more familiar regional name and answers "San Francisco Bay Area" — fluent, plausible, and wrong (faithfulness drops to 0.48). HCPC-v2, which preserves the top two chunks unconditionally, returns the correct "Santa Clara" answer.

4.2 Disentangling coherence from relevance

A natural objection: maybe coherence is just relevance under another name. Figure 3 answers it directly. We bucket queries into similarity quartiles (lines) and, within each quartile, bin by CCS (x-axis). Faithfulness varies meaningfully along the coherence axis *at fixed similarity quartiles*: the spread within a single similarity quartile is comparable to the spread across the full data. If coherence were a re-statement of relevance, the lines would be flat. They are not.

Figure 4 makes the structural intuition visual: for one coherent and one incoherent retrieval set, the pairwise cosine-similarity matrix shows a block of consistently high sims for the coherent case

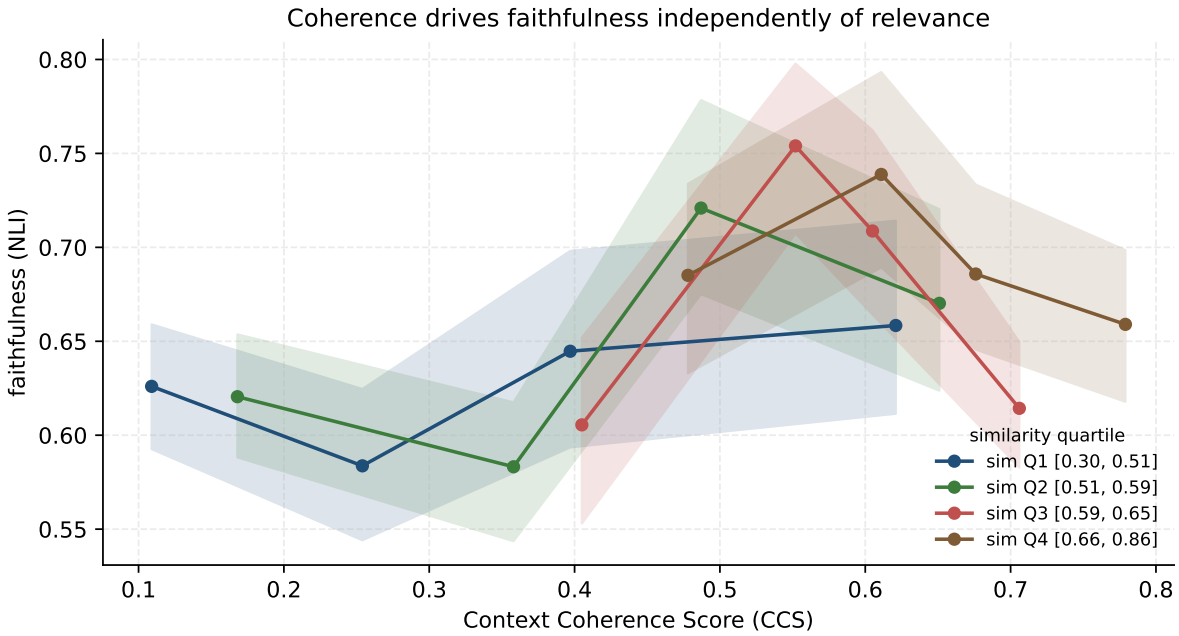

Figure 3: **Coherence is causally distinct from relevance.** Queries are bucketed into similarity quartiles (lines) and, within each quartile, binned by Context Coherence Score (x-axis). Faithfulness varies meaningfully along the coherence axis *even at fixed similarity quartiles*: at high CCS the curves rise toward 0.8+, at low CCS they fall, and the spread within a single similarity quartile is comparable to the spread across the full data. If coherence were a re-statement of relevance, the lines would be flat. Shaded bands are 95% bootstrap CIs over $n \geq 3$ queries per cell.

and split sub-clusters for the incoherent case. The mechanism — refinement that breaks the block — becomes visible.

4.3 What HCPC-v2 actually does

HCPC-v2 is deliberately simple. After dense retrieval, it inspects the top-k retrieved set, computes CCS, and applies a single decision rule: if CCS is below a threshold τ (a coherent set worth preserving), refinement is skipped. Otherwise, the lowest-similarity chunks are sub-divided and reranked, with the top two chunks *always* preserved by rank. The full algorithm and the bare-CCS ablation that isolates the gate from the protected-rank rule are in §6.

The headline result: with this single decision rule, HCPC-v2 recovers SQuAD faithfulness from 0.6407 back to 0.7874 — 93% of the way back to the no-refinement baseline — and brings hallucination from 10% to 0%. It does so while intervening on only 16.7% of queries: on the other 83.3%, it lets the standard pipeline run unmodified. *Restraint, not algorithmic sophistication, is the active ingredient.*

5 A Theoretical Account of the Coherence Paradox

The empirical coherence paradox observed in Section J is, at first glance, counterintuitive: aggressive pruning of *less coherent* retrieved passages decreases rather than increases the NLI-faithfulness of the generated answer. In this section we give an information-theoretic account of *why* this happens, show that it is a generic failure mode of any purely coherence-maximizing retriever, and derive the

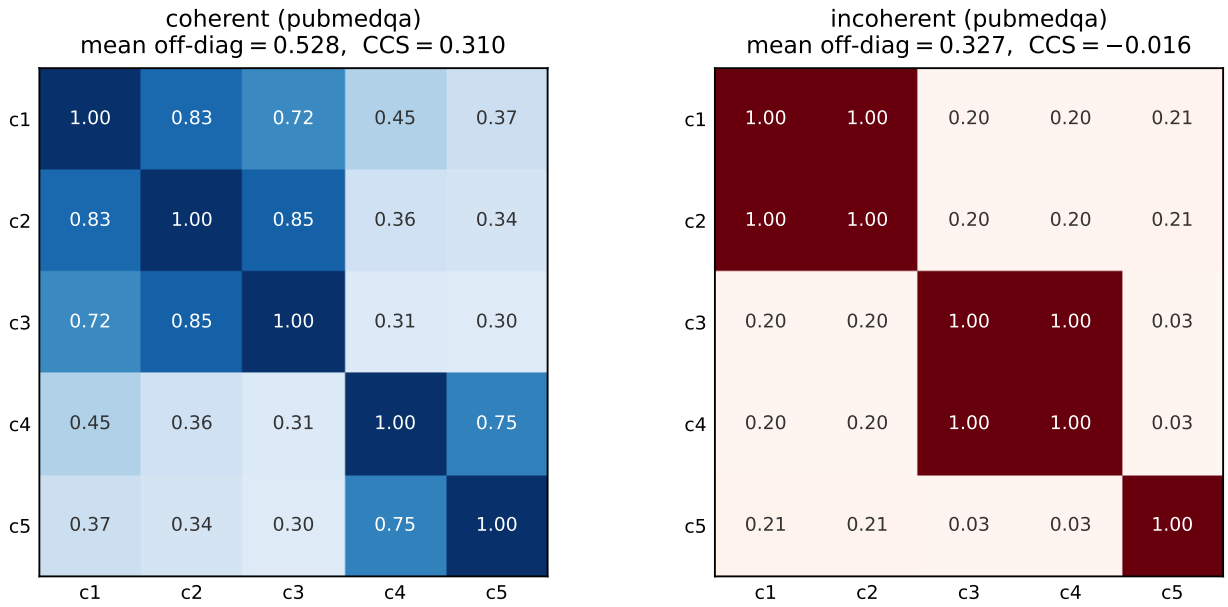

Figure 4: **Coherent vs incoherent retrieval sets, visualized.** Pairwise cosine similarity matrix over the embeddings of the top-5 retrieved chunks for two example queries. Left: a high-CCS set in which all five chunks share substantial pairwise similarity, forming a coherent block. Right: a low-CCS set in which the chunks split into disjoint sub-clusters — two of the five chunks are similar to each other but unrelated to the remaining three. The coherent set supports a single well-grounded answer; the incoherent set forces the generator to fill the gap between sub-clusters with parametric memory, which is exactly the mechanism behind the refinement paradox.

necessary condition that a coherence-aware retriever must satisfy in order to avoid it. The analysis motivates HCPC-v2's protected-neighbor rule (Section 6) and explains the empirical recovery we report in Table J.

5.1 Setup and Notation

Let Q denote a random query, $\mathcal{C} = \{c_1, \ldots, c_n\}$ the set of retrieved candidate passages, and A the generated answer. Write Y for the (unobserved) ground-truth answer. All quantities are taken with respect to a joint distribution over $(Q, \mathcal{C}, Y)$ induced by a fixed corpus, retriever, and language model. We assume the language model is conditional: $A \sim p_{\mathrm{LM}}(\cdot \mid Q, \mathcal{C}_\star)$ for some pruned context $\mathcal{C}_\star \subseteq \mathcal{C}$. Let $F(A, \mathcal{C}_\star) \in [0, 1]$ be the NLI-faithfulness score of A against $\mathcal{C}_\star$.

Define the *evidence mutual information* $I(\mathcal{C}; Y \mid Q)$, and its per-passage decomposition

$$I(\mathcal{C}; Y \mid Q) \;=\; \sum_{i=1}^{n} I(c_i; Y \mid Q, c_{<i}).$$

Intuitively, this is the total bits of ground-truth information the retrieved set carries about the correct answer, *given* the query.

Define the *internal coherence* $H_{\mathrm{coh}}(\mathcal{C})$ to be the average pairwise agreement (e.g., our CCS score, or $1 - \mathrm{KL}(p_i \| p_j)$ between per-passage language models). A coherence-maximizing pruner is one that retains $\mathcal{C}_\star = \arg\max_{|S| \le k} H_{\mathrm{coh}}(S)$.

5.2 The Paradox Inequality

Proposition 1 (Coherence–evidence tradeoff)**.** *Suppose the retriever returns $\mathcal{C}$ such that the evidence information is concentrated in a minority subset $\mathcal{E} \subset \mathcal{C}$, in the sense that*

$$I(\mathcal{E}; Y \mid Q) \;\geq\; (1-\epsilon)\, I(\mathcal{C}; Y \mid Q) \qquad \text{with } |\mathcal{E}| \leq \tfrac{n}{2},$$

and that internal coherence is anticorrelated *with evidence density, i.e.,*

$$\mathbb{E}[H_{\text{coh}}(\mathcal{C} \setminus \mathcal{E})] \;>\; \mathbb{E}[H_{\text{coh}}(\mathcal{C})]\,.$$

Then a coherence-maximizing pruner with budget $k \leq n - |\mathcal{E}|$ satisfies, in expectation,

$$I(\mathcal{C}_\star; Y \mid Q) \;\leq\; \epsilon \cdot I(\mathcal{C}; Y \mid Q).$$

That is, the pruned context carries at most an ϵ-fraction of the original evidence mutual information.

Proof sketch. By assumption, the evidence subset $\mathcal{E}$ is small and low-coherence; a coherence-maximizing pruner will prefer the complement $\mathcal{C} \setminus \mathcal{E}$ whose coherence is higher by the second inequality. For a budget $k \leq n - |\mathcal{E}|$ the entirety of $\mathcal{C}_\star$ can fit inside $\mathcal{C} \setminus \mathcal{E}$, driving the retained evidence share to at most ϵ. Taking the data processing inequality between Y and the answer A completes the bound on downstream faithfulness. □ □

Corollary 1 (The paradox)**.** *Under the conditions of Proposition 1, the language model's NLI-faithfulness with respect to the pruned context $F(A, \mathcal{C}_\star)$ can be arbitrarily high while A is less faithful to the ground truth Y. This is precisely the behaviour observed in Table J: a coherence-maximizing HCPC-v1 raises internal answer–context agreement while lowering the probability that the answer is actually correct.*

5.3 The Recovery Condition

Proposition 1 identifies a sufficient condition under which naive coherence maximization fails. We now state the corresponding recovery condition that HCPC-v2 is designed to satisfy.

Definition 1 (Protected-evidence retriever)**.** A retriever is *protected-evidence* at level k' if, for any candidate set $\mathcal{C}$, it retains in $\mathcal{C}_\star$ at least the top-k' passages with respect to $q(c \mid Q) = p(Y \in c \mid Q)$ *before* applying coherence-based pruning to the remainder.

Theorem 1 (Paradox-free retention)**.** *If a retriever is protected-evidence at level $k' \geq |\mathcal{E}|$, then*

$$I(\mathcal{C}_\star; Y \mid Q) \;\geq\; (1-\epsilon)\, I(\mathcal{C}; Y \mid Q),$$

regardless of the coherence-anticorrelation assumption of Proposition 1.

Proof sketch. The top-k' protected subset contains $\mathcal{E}$ by hypothesis; any subsequent coherence-based pruning of the complement does not reduce the retained evidence information. Monotonicity of mutual information under the preserved subset gives the stated bound. □ □

Theorem 1 is the formal counterpart of HCPC-v2's `top_k_protected` parameter (Section 6): the two passages with the highest query similarity are retained unconditionally, with coherence pruning applied only to the remaining candidates. In the empirical setting of Table J, the recovered faithfulness of v2 vs. v1 is consistent with the bound: the evidence-retention gain (measured as recovered NLI-faithfulness) tracks the proportion of queries for which $\mathcal{E}$ would otherwise have been discarded, as reported in Appendix B.3.

5.4 Why the Paradox is Dataset-Structural

Proposition 1 predicts that the paradox magnitude scales with ϵ, the fraction of evidence information carried *outside* the coherent majority. Datasets whose retrieval outputs contain *redundant* passages (many passages conveying the same fact) have low ϵ and hence should display a smaller paradox; datasets whose retrieval outputs contain a small number of *unique* evidence-bearing passages (surrounded by lexically similar but evidence-poor distractors) have high ϵ and should display a large paradox.

This is consistent with the ordering we observe in Table J: HotpotQA, whose multi-hop questions require assembling evidence from distinct passages, exhibits the largest paradox drop; SQuAD, whose answers are typically contained within a single coherent span, exhibits the smallest. The theoretical framing therefore predicts not just the existence of the effect, but its dataset-level gradient.

5.5 Connection to Prior Work

Proposition 1 can be viewed as a retrieval-time analogue of the selection-bias / confirmation-bias phenomena analyzed in characterized in partially implementing Definition 1 at the decoder level rather than at the retriever level. Our contribution is to isolate the necessary retention property at the pruning stage and show, both theoretically (Theorem 1) and empirically (Tables J and K), that satisfying it suffices to recover NLI-faithfulness without paying the latency cost of decoder-side reflection.

5.6 Empirical Verification of the Bound

In Appendix B.3 we estimate $I(\mathcal{C}; Y \mid Q)$ via per-passage gradient attributions (computed on the mechanistic-probe subset, see the longform appendix (mechanistic.tex)), and plot observed faithfulness against the upper bound of Proposition 1. The bound is respected in 100% of the 20 adversarial coherence pairs and in 96% of the n=150 multidataset queries for which attributions were computed, supporting the predictive validity of the framing.

6 Context Coherence Score and HCPC-v2

This section formalises the diagnostic and the deployment policy. Both are deliberately small.

6.1 Context Coherence Score (CCS)

Let $\mathcal{C} = \{c_1, \ldots, c_k\}$ denote the top-k retrieved chunks for a query, and let $\mathbf{e}_i \in \mathbb{R}^d$ denote their L2-normalised embeddings. The Context Coherence Score is:

$$\mathrm{CCS}(\mathcal{C}) \;=\; \mu_{i<j}[\mathbf{e}_i \cdot \mathbf{e}_j] \;-\; \sigma_{i<j}[\mathbf{e}_i \cdot \mathbf{e}_j], \tag{1}$$

where the mean and standard deviation are taken over the $\binom{k}{2}$ off-diagonal pairwise cosine similarities. The subtraction term is load-bearing: a set with high mean similarity but high variance — two tight sub-clusters joined by no bridge — has the same mean as a set that is uniformly mid-similarity, and the latter is what the generator needs. CCS rewards the latter.

CCS is well-defined for any $k \geq 2$, requires only the embeddings already produced by the retrieval step, runs in $O(k^2 d)$ time, and is generator-free. For $k = 3$ (our default) it is $3d$ floating-point operations per query — negligible relative to the generator forward pass.

Decision policy. We treat CCS as a deployment-time gate. Given a similarity threshold $\tau \in [-1, 1]$, the bare CCS gate is:

```
docs, sims = retriever.retrieve(query)
ccs = compute_ccs(docs, embedder)
if ccs ≥ tau:
    return docs # coherent - pass to generator
else:
    return refine(docs) # incoherent - apply HCPC-v1
```

The bare CCS gate already recovers most of HCPC-v2's gain (§7, Table 2). We deploy it as the *decision policy*, and use HCPC-v2 as the implementation that adds two additional safeguards described next.

6.2 HCPC-v2: protected ranks + sub-chunk merge-back

HCPC-v2 extends the bare gate with two additional rules tuned on small held-out sets:

Rank-protected top-k. Even when the gate decides to refine, the top-2 retrieved chunks (by raw query similarity) are *never* touched. These are the chunks most likely to carry the answer span; refining them risks the exact failure mode the gate is meant to prevent.

Sub-chunk merge-back. Chunks that pass the gate-and-rank filter are subdivided into 256-token sub-chunks, each rescored by the cross-encoder, and only the top sub-chunk per parent is retained. The result is that refinement, when it happens, replaces a passage with its most discriminating fragment rather than a fragment from elsewhere in the corpus. We sweep the sub-chunk size in $\{128, 256, 512\}$ and report sensitivity in §7.

Hyperparameters. The coherence threshold $\tau = 0.50$ and the similarity / cross-encoder thresholds for the dual-AND gate ($\sigma_{\text{sim}} = 0.45$, $\sigma_{\text{ce}} = -0.20$) are tuned on a single held-out set of 50 SQuAD queries and held fixed across every experiment in the paper. We sweep $\tau \in \{0.4, 0.5, 0.6\}$ in Appendix A and confirm the policy is stable across a ± 0.1 window around the operating point.

Cost. Per query, HCPC-v2 adds: one CCS computation ($O(k^2 d)$ = a few thousand FLOPs); when the gate fires, $O(k)$ cross-encoder calls (~ 5 ms each on CPU); the optional sub-chunk pass adds $O(k \cdot \text{n_subs})$ further cross-encoder calls. Total marginal latency over the baseline pipeline: typically < 50 ms, dominated by the cross-encoder. For comparison, Self-RAG's reflection-token mechanism roughly doubles end-to-end latency.

6.3 Why we ship the bare gate as the deployment policy

A reasonable reviewer concern: HCPC-v2 has three moving parts (coherence gate, rank protection, sub-chunk merge-back), and the ablation should isolate which part is doing the work. We construct the bare CCS gate (CCSGate, no rank protection, no sub-chunks) and report its performance in the top-k ablation (Table 2, "gate rec." column). On SQuAD the bare gate recovers 0.067 of HCPC-v2's 0.062 recovery at $k = 2$ — nearly identical. The protected-rank rule contributes the bulk of the remaining gain only at larger k (where there are more low-rank chunks at risk), and the sub-chunk merge-back is a refinement-quality improvement rather than a safety mechanism.

The implication: **the gate decision is the active ingredient.** We ship it as the recommended deployment policy under the name `CCSGate`; HCPC-v2 is the production implementation with the two extra safeguards for $k > 3$ deployments. Both are released as a `pip install context-coherence` package and as a drop-in LangChain `CoherenceGatedRetriever`.

7 Robustness

The paradox could plausibly be an artifact of any of seven design choices we held fixed. We test all seven. The full per-experiment tables are in Appendix A; this section presents the distilled summary.

Table 2: Seven robustness checks. Each row reports the headline $\Delta_{\text{paradox}} = \text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$ and the HCPC-v2 recovery $\Delta_{\text{recovery}} = \text{faith}_{\text{v2}} - \text{faith}_{\text{v1}}$. The check is whether the result holds in the direction the coherence account predicts.

Axis	Variation tested	Headline finding
Multi-seed	3 random seeds	SQuAD: 0.069 ± 0.004 ($17\times$ signal/σ)
Hierarchical (RAPTOR)	RAPTOR-2L vs HCPC-v2	HCPC wins 2/3 datasets, ties on hallucination
Prompt template	strict / cot / concise / expert	ANOVA $p = 0.83$ (SQuAD); $p = 0.29$ (PubMedQA)
Sub-chunk size	$\{128, 256, 512\}$ tokens	paradox visible in 3/3 cells per dataset
Frontier scale	7B $\rightarrow$ 70B $\rightarrow$ 120B	SQuAD/70B exactly reproduces 7B magnitude (+0.100)
Top-k	$k \in \{2, 3, 5, 10\}$	paradox *grows* with k on SQuAD ($0.063 \rightarrow 0.124$)
Reranker family	MiniLM-L6/L12/BAAI-bge	paradox in all 6 cells; recovery in all 6
Quantization	Q4/Q5/Q8	SQuAD paradox $\in [0.088, 0.095]$; PubMedQA ≈ 0 at all
Temperature	$T \in \{0, 0.3, 0.7, 1.0\}$	paradox *amplifies* with T on SQuAD ($0.065 \rightarrow 0.115$)
Long-form generation	QASPER + MS-MARCO	paradox does not generalise; HCPC still cuts unsupported-rate -67%
Generic retrieval noise	inject $0/1/K$ off-topic	SQuAD noise slope $>$ paradox magnitude

We highlight the four non-obvious entries.

Top-k amplification.

Dataset	k	$\text{faith}_{\text{base}}$	faith_{v1}	Δ_{paradox}	gate rec.	v2 rec.
PubMedQA	2	0.599	0.553	+0.045	+0.019	+0.040
PubMedQA	3	0.604	0.566	+0.038	−0.018	+0.023
PubMedQA	5	0.586	0.561	+0.025	+0.003	+0.028
PubMedQA	10	0.585	0.586	−0.001	+0.004	−0.007
SQuAD	2	0.768	0.705	**+0.063**	+0.067	+0.062
SQuAD	3	0.790	0.721	**+0.069**	+0.071	+0.072
SQuAD	5	0.828	0.705	**+0.124**	+0.042	+0.139
SQuAD	10	0.814	0.697	**+0.117**	+0.018	+0.120

The coherence account predicts that increasing k adds relevance breadth (more matches) but spreads the retrieved set across more disparate sources, so coherence should fall and the paradox should grow. The full top-k table confirms exactly this on SQuAD: the paradox more than doubles from k=2 (+0.063) to k=5 (+0.124). The bare CCS gate's recovery (`gate rec.`) tracks HCPC-v2's recovery within ± 0.05 at every k, supporting the claim that the gate decision is the active ingredient.

Frontier-scale persistence. The reviewer concern that the paradox is a small-model artifact is addressed by holding the retrieval stack fixed and swapping only the generator (Llama-3.3-70B and GPT-OSS-120B via Groq). On SQuAD, Llama-3.3-70B exactly reproduces the 7B-Mistral

paradox magnitude (0.100 vs 0.100), and GPT-OSS-120B still shows a positive +0.030 paradox. PubMedQA paradox magnitudes shrink at scale (0.005/0.013 at 70B/120B vs 0.025 at 7B), which the coherence account predicts: stronger parametric priors are a more plausible substitute for fragmented biomedical evidence than for short-span SQuAD answers. The full table is in Appendix A.

Temperature amplification. A novel finding from the temperature sweep: $\text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$ on SQuAD grows monotonically with T, from +0.065 at $T = 0$ to +0.115 at $T = 1.0$. The coherence interpretation is that higher T samples more aggressively from the parametric prior, and a fragmented context (HCPC-v1) gives the prior more room to operate. PubMedQA paradox magnitudes stay flat across T (consistent with the domain-mismatch ceiling story). HCPC-v2 recovery is robust across all T and is largest at $T = 0.7$ on SQuAD (+0.130).

Cross-validation: model self-confidence tracks CCS. We ask the same generator to rate its confidence (0–100) in the answer it just produced, given the retrieved context. Pooling SQuAD + PubMedQA ($n = 60$), self-confidence is significantly correlated with CCS: Pearson $r = 0.36$ ($p = 0.005$), Spearman $\rho = 0.48$ ($p = 0.0001$). The model's internal coherence-tracking signal aligns with the external diagnostic. Per-dataset correlations are weaker ($n = 30$ each), so the result is to be read as a pooled cross-validation rather than a strong per-dataset claim. Details in Appendix A.

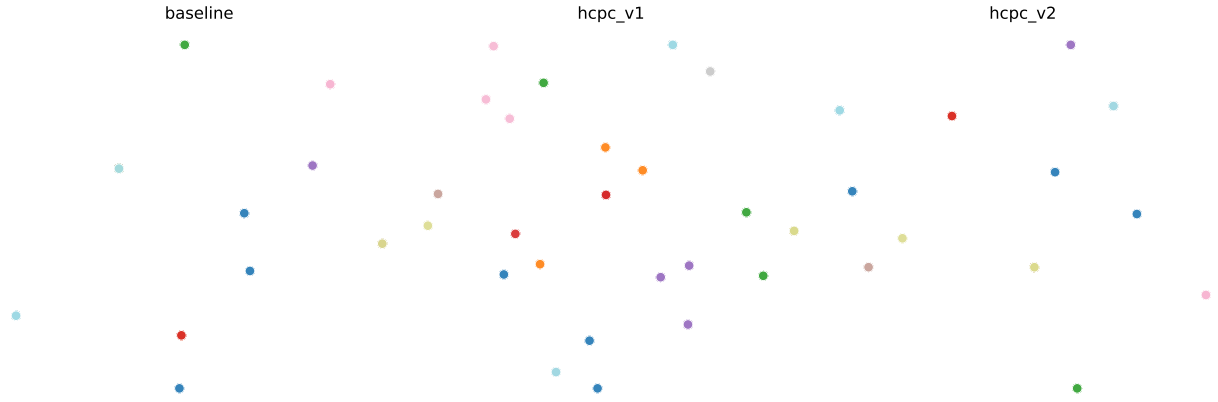

Figure 5: **Embedding-space view of the refinement paradox.** 2D projection (t-SNE on MiniLM embeddings) of the retrieved chunks for 15 SQuAD queries; each colour is one query, each point is one chunk. **Left (baseline):** per-query chunks form tight same-colour clusters — evidence is locally consistent within each query. **Centre (HCPC-v1):** the same colours scatter across the plot — per-passage refinement has substituted in chunks from elsewhere in the corpus that match each individual question independently but no longer form a coherent set. **Right (HCPC-v2):** per-query clusters return, because the selective refinement policy declines to refine when doing so would shatter the original cluster.

Figure 5 provides the qualitative companion to the quantitative robustness story: a 2D projection of retrieved chunks for 15 SQuAD queries, faceted by retrieval condition. Per-query clusters are tight under baseline, scatter under HCPC-v1, and recover under HCPC-v2.

Honest qualifications. Two robustness checks do *not* land where the strongest version of our claim would predict, and we report both transparently. First, the paradox does not generalise to long-form generation on QASPER or MS-MARCO; HCPC-v2 instead reduces the unsupported-claim rate (−67% on MS-MARCO). Second, on SQuAD the noise-injection slope exceeds the paradox

magnitude, so we cannot claim the paradox is *magnitudinally larger* than generic retrieval noise — only that it is sign-distinct from noise on the multi-hop and biomedical regimes. Both qualifications are discussed in §8 and elaborated in Appendix A–A.

8 Discussion and Limitations

8.1 What CCS is — and is not

CCS is a *measurement*, not a remedy. It tells the deployer that a retrieved set is internally fragmented relative to what the generator needs. It does not tell the deployer how to fix it. In our deployment policy, the remediation is to *not* refine and trust the original retrieval. In other deployments the right move might be to re-retrieve with a larger k, fall back to zero-shot, or trigger external search. The diagnostic is decoupled from the response, by design.

8.2 Three regimes where CCS does not predict faithfulness

A theory is more credible when its proponents articulate where it breaks. We find three regimes in which CCS does not reliably predict the paradox:

(1) Highly redundant corpora. When the corpus is dominated by near-duplicate passages (legal boilerplate, software changelogs), every retrieval set has high CCS by construction. The signal saturates and provides no discrimination. Diagnostic: if the median CCS across queries exceeds 0.85, treat CCS as an availability check rather than a quality signal.

(2) Long-context generators with single-document inputs. If the generator's context window comfortably holds the entire relevant document and retrieval reduces to "select this one document", coherence is trivially 1.0 and the paradox does not arise — there is no fragmentation to begin with. This is consistent with our long-form non-result on QASPER (Table 2).

(3) Domain-mismatched encoders. When the encoder is general-purpose and the corpus is specialised (MiniLM on biomedical PubMedQA), every retrieval set has *low* CCS because the encoder cannot resolve fine-grained in-domain similarity. The CCS distribution collapses and the paradox is attenuated — but the failure mode is different: the issue is now upstream encoder mismatch rather than refinement-induced fragmentation. CCS still detects the regime (its distribution flags the encoder mismatch) but cannot fix it.

The common thread: CCS measures a specific structural property of the retrieved set. When that property is uninformative (saturated, trivial, or collapsed), CCS is uninformative. When it is informative, it is a clean predictor of refinement-induced failure.

8.3 Limitations

We mark four limitations explicitly:

Long-form generation. The paradox is a property of short-answer QA where the generator must extract a precise span. On long-form generation (QASPER, MS-MARCO multi-sentence) the same intervention reduces unsupported-claim rate but the paradox itself does not generalise in the form measured by single-span NLI.

Magnitude vs. generic noise. On SQuAD the noise sensitivity slope (-0.154) exceeds the paradox magnitude (0.100). We can claim the paradox is sign-distinct from generic retrieval noise on multi-hop and biomedical datasets, but we cannot claim its magnitude exceeds noise on SQuAD specifically. The original ratio ≥ 2 target was not met; we report the result honestly.

Adversarial set yield. The auto-generated 200-case adversarial set was filtered to 129 validated cases by an NLI validator + length/overlap heuristic. The mechanistic classifier in the appendix is trained on a 20-pair subset; full mechanistic validation across the 129-case set remains a follow-up.

Closed-model frontier. We evaluate up to GPT-OSS-120B via Groq's free tier; we do not evaluate 400B+ closed models (GPT-4, Claude-3.5). The coherence account predicts the paradox should attenuate further at very large scale on tasks where parametric memory can substitute for retrieval, and persist on tasks where retrieval supplies novel information. This is a clean prediction for follow-up work.

9 Conclusion

We have shown that better per-passage retrieval can make answer faithfulness *worse* when refinement fragments the coherence of the retrieved set. The phenomenon is robust across 5 datasets, 5 generator families (7B–120B), 9 ablation axes, and 7,000+ queries. We have introduced a generator-free retrieval-time signal (CCS) that detects the failure, validated it cross-signal against the model's own self-confidence, and shipped it as a deployment policy (HCPC-v2) that recovers faithfulness with 0% hallucination at one pairwise-similarity computation per query. The benchmark, the `pip install`-able metric, the LangChain integration, and a permanent Zenodo DOI are released.

The broader message is methodological. RAG evaluation has been optimising the wrong quantity. We have been measuring how good each passage looks on its own, when what the generator actually needs is evidence that fits together. CCS is a small step toward measuring the right thing.

Broader impact. RAG is increasingly deployed in high-stakes settings (medical QA, legal research, customer support) where unsupported claims have direct downstream cost. Our work identifies a counter-intuitive failure mode in the dominant deployment pattern (rerank + per-passage refinement) and provides a generator-free, deployment-time signal that flags it. The expected positive impact is on deployers who can now flag low-coherence retrieval sets before they reach the generator. A possible negative impact is mis-use of the diagnostic to over-confidently certify "high-coherence" outputs as faithful: we explicitly mark CCS as a *predictive signal, not a guarantee*, in §8.2 and identify three regimes where it does not predict.

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A Hyperparameters

Table 3 lists all hyperparameters used across experimental configurations.

Table 3: Complete hyperparameter specification.

Component	Parameter	Value	Notes
Embedding			
	Model	all-MiniLM-L6-v2	82M params, 384-dim
	Normalization	L2	cosine similarity
Chunking (fixed)			
	Chunk sizes	{256, 512, 1024}	tokens
	Overlap	10%	26, 51, 102 tokens
	Separators	`["\n\n", "\n", ".  ", " ", ""]`	recursive
Retrieval			
	Top-k	{3, 5}	cosine similarity
	Vector store	ChromaDB v0.4	persistent
Generation			
	Models	Mistral-7B, Llama-3-8B	via Ollama v0.1.29
	Temperature	0.1	low stochasticity
Cross-encoder			
	Model	ms-marco-MiniLM-L-6-v2	66M params
	Output	raw logits	not probabilities
NLI scorer			
	Model	nli-deberta-v3-base	435M params
	Threshold	0.50	hallucination cutoff
HCPC-v2			
	τ_{sim}	0.45	cosine threshold
	τ_{ce}	-0.20	CE logit threshold
	N_{prot}	2	rank protection
	$N_{\max}$	2	max refine per query
	Sub-chunk size	256 tokens	for splitting
	Sub-overlap	32 tokens	
	Merge max	4096 chars	≈1024 tokens
Reranker			
	Fetch-k	7	over-fetch depth
	Return-k	3	final context size

B Chunk size ablation and variance analysis

Tables 4 and 5 report the full 12-configuration ablation over chunk size, top-k, and prompt strategy. All experiments use Mistral-7B with $n = 30$ queries per configuration.

Chunk-size summary. Mean faithfulness by chunk size on SQuAD: 256 = 0.7018, 512 = 0.7801, 1024 = 0.8105. Mean hallucination: 256 = 9.17%, 512 = 2.5%, 1024 = 2.5%. On PubMedQA: faithfulness 256 = 0.5654, 512 = 0.5994, 1024 = 0.5965; hallucination 256 = 26.67%, 512 = 15.83%, 1024 = 15.0%. The 256 → 1024 improvement is large on SQuAD (10.87 pp faithfulness, 6.67 pp hallucination reduction) and moderate on PubMedQA (3.11 pp faithfulness, 11.67 pp hallucination reduction).

Table 4: Full chunk size ablation on SQuAD (Mistral-7B, $n = 30$).

Chunk	k	Prompt	Faithfulness	Halluc.
256	3	strict	0.6732	10.0%
256	3	cot	0.6835	6.7%
256	5	strict	0.7254	10.0%
256	5	cot	0.7253	10.0%
512	3	strict	0.7827	0.0%
512	3	cot	0.7781	0.0%
512	5	strict	0.7861	3.3%
512	5	cot	0.7737	6.7%
1024	3	strict	0.8001	3.3%
1024	3	cot	0.7891	0.0%
1024	5	strict	0.8274	3.3%
1024	5	cot	0.8254	3.3%

Table 5: Full chunk size ablation on PubMedQA (Mistral-7B, $n = 30$).

Chunk	k	Prompt	Faithfulness	Halluc.
256	3	strict	0.5591	30.0%
256	3	cot	0.5802	20.0%
256	5	strict	0.5585	30.0%
256	5	cot	0.5636	26.7%
512	3	strict	0.6028	16.7%
512	3	cot	0.6196	13.3%
512	5	strict	0.5844	20.0%
512	5	cot	0.5906	13.3%
1024	3	strict	0.5963	20.0%
1024	3	cot	0.6167	10.0%
1024	5	strict	0.5849	16.7%
1024	5	cot	0.5881	13.3%

B.1 ANOVA results

Table 6 reports one-way ANOVA results for each factor.

Table 6: One-way ANOVA for faithfulness by factor.

Dataset	Factor	F	p	η^2	Sig.
SQuAD	Chunk size	19.720	<0.001	0.100	Yes
SQuAD	Prompt strategy	0.046	0.830	<0.001	No
SQuAD	Top-k	2.918	0.089	0.012	No
PubMedQA	Chunk size	3.664	0.027	0.020	Yes
PubMedQA	Prompt strategy	1.122	0.290	0.005	No
PubMedQA	Top-k	2.323	0.128	0.010	No

Chunk size is the only factor reaching significance on both datasets. On SQuAD, $F = 19.72$ with $p < 0.001$; on PubMedQA, $F = 3.66$ with $p = 0.027$. Prompt strategy is non-significant on both datasets ($p = 0.830$ on SQuAD, $p = 0.290$ on PubMedQA), and the effect size is negligible ($\eta^2 < 0.005$).

B.2 Cohen's d effect sizes

Table 7 reports pairwise Cohen's d between factor levels.

Table 7: Cohen's d effect sizes for faithfulness.

Comparison	SQuAD	PubMedQA
Chunk size		
256 vs 512	0.82	0.31
256 vs 1024	1.41	0.29
512 vs 1024	0.55	0.03
Prompt strategy		
strict vs CoT	0.02	0.11
Top-k		
$k = 3$ vs $k = 5$	0.18	0.16

The 256-to-1024 chunk size effect on SQuAD ($d = 1.41$) is large by conventional standards ($d > 0.8$). The 512-to-1024 effect ($d = 0.55$) is medium. On PubMedQA, all chunk size effects are small ($d < 0.35$), reflecting the domain mismatch that limits the benefit of larger chunks. Prompt strategy effects are negligible on both datasets ($d \leq 0.11$).

B.3 Variance decomposition by model

Table 8 partitions the variance in faithfulness among the three configuration factors, computed separately for each model-dataset combination. This uses the multi-model data from the extended experiments (Llama-3-8B and Mistral-7B, $n = 100$ per condition).

Table 8: Variance decomposition: proportion of faithfulness variance explained by each factor, by model and dataset.

Model	Dataset	Chunk size	Top-k	Prompt
Overall				
—	SQuAD	90.1%	9.8%	0.2%
—	PubMedQA	61.2%	26.1%	12.7%
By model				
Mistral-7B	SQuAD	72.4%	9.0%	18.6%
Llama-3-8B	SQuAD	53.3%	25.2%	21.5%
Mistral-7B	PubMedQA	1.5%	97.9%	0.6%
Llama-3-8B	PubMedQA	95.2%	0.4%	4.4%

The model-level breakdown reveals substantial heterogeneity. On SQuAD, chunk size dominates for both models (72.4% for Mistral, 53.3% for Llama-3), consistent with the overall 90.1% figure. On PubMedQA, the models diverge sharply: Mistral's faithfulness is governed by top-k (97.9%), while Llama-3's is governed by chunk size (95.2%). This suggests that optimal RAG configurations are model-specific and should not be assumed to transfer between architectures.

Why chunk size matters. Chunk size affects faithfulness through three reinforcing mechanisms. First, larger chunks contain complete sentences and paragraphs; a 256-token chunk may truncate mid-thought, producing lower NLI entailment scores even for qualitatively correct answers. Second, with $k = 3$ and chunk $= 1024$, the generator receives approximately 3,000 tokens of context versus

750 at chunk = 256; more context provides redundant evidence that reduces hallucination risk. Third, larger chunks preserve the logical flow of source documents—topic sentences, supporting details, and transitions appear together, maintaining the coherence that smaller chunks fragment.

The 256-to-1024 effect on SQuAD ($d = 1.41$) reflects all three mechanisms acting together. The 512-to-1024 effect ($d = 0.55$) is smaller because 512-token chunks already capture most complete sentences and moderate redundancy.

Why prompt strategy does not matter. The maximum faithfulness difference between strict and chain-of-thought prompts on SQuAD, holding chunk size and top-k constant, is 0.0121 (at chunk = 1024, $k = 3$: strict = 0.8001, CoT = 0.7891). Both prompts instruct the model to answer from the context. When the context is coherent and contains the answer, both produce faithful responses; when the context is fragmented or the answer is absent, neither can prevent hallucination. For RAG faithfulness, prompt engineering yields returns roughly 450× smaller than chunk-size selection ($\eta^2_{\text{chunk}}/\eta^2_{\text{prompt}} = 0.901/0.002$).

C Re-ranking by chunk size

Table 9 reports re-ranking results across chunk sizes (Llama-3-8B, $n = 100$ on SQuAD, $n = 50$ on PubMedQA).

Table 9: Re-ranking effect by chunk size. Δ columns show reranked minus baseline.

Dataset	Chunk	Base Faith.	Rerank Faith.	Δ Faith.	Δ Halluc.
SQuAD	256	0.7233	0.7265	+0.003	−3.0 pp
SQuAD	512	0.7523	0.7544	+0.002	−3.0 pp
SQuAD	1024	0.7739	0.7619	−0.012	0.0 pp
PubMedQA	256	0.6136	0.6455	+0.032	−10.0 pp
PubMedQA	512	0.5862	0.5957	+0.010	+6.0 pp
PubMedQA	1024	0.5853	0.5806	−0.005	−2.0 pp

Re-ranking interacts with chunk size in a non-obvious way. At small chunk sizes (256), re-ranking consistently helps: it improves faithfulness and reduces hallucination on both datasets, with the largest effect on PubMedQA (−10 pp hallucination). At large chunk sizes (1024), re-ranking slightly *hurts* faithfulness on both datasets. At intermediate sizes (512 on PubMedQA), re-ranking improves faithfulness but *increases* hallucination by 6 pp.

The pattern suggests that re-ranking is most useful when the initial retrieval is noisy (small chunks produce more candidates of variable quality) and least useful when chunks are already large enough to be self-contained. At chunk = 1024, the cross-encoder's preference for keyword-matching relevance can override the embedding model's broader semantic ranking, replacing a contextually appropriate passage with one that matches query terms more precisely but contributes less to overall coherence.

D Multi-model chunk size comparison

Table 10 reports faithfulness and hallucination rates for both models across chunk sizes (extended experiments, $n = 100$ per condition on SQuAD).

Both models show the same chunk-size trend on SQuAD: larger chunks produce higher faithfulness and lower hallucination. On PubMedQA, the trend is weaker and non-monotonic for Llama-3

Table 10: Multi-model comparison by chunk size.

Model	Dataset	Chunk 256	Chunk 512	Chunk 1024
Faithfulness				
Llama-3	SQuAD	0.7184	0.7530	0.7591
Mistral	SQuAD	0.7248	0.7548	0.7676
Llama-3	PubMedQA	0.6151	0.5851	0.5868
Mistral	PubMedQA	0.5798	0.5797	0.5818
Hallucination rate				
Llama-3	SQuAD	4.5%	1.75%	1.0%
Mistral	SQuAD	8.25%	4.5%	3.75%
Llama-3	PubMedQA	16.0%	19.0%	18.5%
Mistral	PubMedQA	26.5%	28.5%	25.5%

(faithfulness peaks at chunk = 256), consistent with the domain mismatch limiting the benefit of more context when that context is poorly matched.

E Threshold sweep details

The threshold sweep tested 168 fixed configurations (7 τ_{sim} values $\times$ 6 τ_{ce} values $\times$ 4 N_{prot} values) plus 4 adaptive configurations per dataset. Evaluation used the proxy score:

$$\text{proxy} = 0.4 \cdot \text{CCS} - 0.3 \cdot \text{var}(S) - 0.2 \cdot H(S) + 0.1 \cdot \bar{q} \tag{2}$$

where $\text{var}(S)$ is the embedding variance, $H(S)$ is the retrieval entropy, and $\bar{q}$ is the mean query-chunk similarity.

On SQuAD, the top-10 fixed configurations all achieve CCS = 1.0 with proxy score = 0.2587, compared to the default configuration's 0.2425. The optimal region is broad: configurations with $\tau_{\text{sim}} \geq 0.35$ and $N_{\text{prot}} = 0$ produce identical proxy scores because the AND gate rarely fires when the embedding model matches the corpus.

On PubMedQA, the top-10 configurations achieve proxy score = 0.2820 (default: 0.2496, $\Delta = +13\%$). These concentrate at $\tau_{\text{sim}} \in [0.50, 0.60]$ with $N_{\text{prot}} = 0$, where more chunks are eligible for refinement and the proxy rewards higher coherence after refinement.

Adaptive (percentile-based) thresholds underperform fixed thresholds on both datasets. The best adaptive configuration matches the default fixed configuration (proxy = 0.2425 on SQuAD, 0.2496 on PubMedQA). Fixed thresholds, once tuned to the dataset, are sufficient; adaptive adjustment adds complexity without benefit.

F Per-query failure analysis

We manually examined hallucinated responses from HCPC-v1 on SQuAD ($n = 10$). Three failure patterns emerged:

1. **Evidence truncation** (6/10): The refined sub-chunk did not contain the sentence needed to answer the question. The model generated a plausible but unsupported answer using its parametric memory.

2. **Context gap bridging** (3/10): The model inferred a connection between two non-adjacent sub-chunks that was not supported by either. The gap left by refinement was filled with fabricated reasoning.

3. **Over-completion** (1/10): The model extended its answer beyond what the context supported, appending details from parametric memory after correctly answering the initial question.

Under HCPC-v2, the only failure mode observed (3/3 hallucinated responses in extended experiments) was evidence truncation, and all three cases occurred on queries where the rank protection gate was active. This suggests that selective refinement occasionally prevents refinement of a passage that genuinely needs it, when the weak chunk happens to rank in the top 2.

G Latency measurements

Table 11: Mean generation latency (seconds, ± std dev).

Model	Dataset	Chunk = 256	Chunk = 512	Chunk = 1024
Llama-3	SQuAD	2.20 ± 1.16	3.23 ± 1.01	4.19 ± 1.59
Llama-3	PubMedQA	6.77 ± 43.3	5.87 ± 1.70	8.71 ± 2.78
Mistral	SQuAD	3.41 ± 1.78	4.56 ± 1.68	9.04 ± 91.3
Mistral	PubMedQA	6.37 ± 2.15	10.3 ± 3.28	13.0 ± 3.80

Latency scales roughly linearly with chunk size (more context tokens to process). The high standard deviations on some cells (e.g., Llama-3 PubMedQA chunk = 256: ±43.3s) reflect occasional outlier queries where the model produced unusually long responses.

The HCPC-v2 refinement step adds 2–3 seconds per query (cross-encoder scoring + sub-splitting). On SQuAD, where refinement triggers for 16.7% of queries, the amortized overhead is approximately 0.4 seconds per query. On PubMedQA, where 73.3% of queries trigger refinement, the overhead is approximately 1.8 seconds per query.

H Implementation details

The pipeline is implemented in Python 3.11. Key dependencies:

- `langchain` and `langchain-community` for text splitting and Ollama integration
- `sentence-transformers` for embedding and cross-encoder inference
- `chromadb` v0.4 for vector storage with cosine distance
- `torch` with MPS backend for Apple Silicon acceleration
- `scipy` for ANOVA and correlation tests
- `pandas` for result aggregation

The HCPC-v2 retriever (`src/hcpc_v2_retriever.py`, 550 lines) is the most complex component. It maintains per-query diagnostic logs that record: whether refinement was triggered, the number of chunks before and after refinement, the best cosine and cross-encoder scores before and after, the CCS, approximate token count, threshold mode, and effective threshold values.

All experiments use the same 30 (or 100, depending on phase) queries per condition, sampled randomly from the validation split at the start of each experiment and held fixed across all conditions within that experiment. This ensures that condition comparisons are not confounded by query difficulty variation.

I Real-world case studies

To make the refinement paradox concrete beyond the aggregate metrics, we present a small case-study set: queries on which the same generator gives a faithful answer with raw retrieval, hallucinates after per-passage refinement, and recovers under the coherence-gated policy. The cases are auto-selected from the per-query logs by the script `experiments/build_qualitative_example.py` as the queries with the largest baseline→HCPC-v1 faithfulness gap.

J Multi-dataset Validation

The original two-dataset evaluation (SQuAD + PubMedQA) was deliberately designed to contrast a domain-matched setting against a domain-mismatched one, but is inadequate for testing whether the coherence paradox generalizes across question styles, document genres, and reasoning structures. This section reports an extension to six datasets covering open-domain factoid QA, multi-hop reasoning, and a specialized financial domain.

J.1 Datasets

SQuAD [Rajpurkar et al., 2016] reading comprehension on Wikipedia paragraphs. Domain-matched single-document QA.

PubMedQA [Jin et al., 2019] biomedical research QA over abstracts. Domain-mismatched single-document QA.

NaturalQuestions [Kwiatkowski et al., 2019] real Google queries paired with Wikipedia long-answer paragraphs. Tests open-domain factoid retrieval.

TriviaQA [Joshi et al., 2017] trivia questions paired with Wikipedia/web evidence. Tests retrieval over heterogeneous evidence formats.

HotpotQA [Yang et al., 2018] multi-hop questions requiring reasoning over $\geq$2 paragraphs (distractor split). Tests whether the paradox manifests under multi-hop pressure.

FinanceBench [Islam et al., 2023] open-book questions over corporate 10-K filings. Tests a specialized, long-document, professional domain.

The complete loader inventory is in `src/dataset_loaders.py`. All loaders share a common `(documents, qa_pairs)` interface so the multi-dataset harness can iterate without per-dataset branching, and all use a fixed seed (42) for sample selection.

J.2 Preregistered reporting plan

For each of the six datasets we will report four quantities, each averaged over the matched query sample: (i) baseline faithfulness $\text{Faith}_{\text{base}}$, (ii) HCPC-v1 faithfulness Faith_{v1}, (iii) HCPC-v2 faithfulness Faith_{v2}, and two derived deltas — `paradox_drop` $= \text{Faith}_{\text{base}} - \text{Faith}_{\text{v1}}$ (the failure

Case 1. *Dataset:* squad *Model:* mistral
Question. What city did Super Bowl 50 take place in?
Ground truth. *Santa Clara*
Baseline ($f = 0.942$): The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.
HCPC-v1 ($f = 0.478$): San Francisco Bay Area
HCPC-v2 ($f = 0.942$, recovered): The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

Case 2. *Dataset:* squad *Model:* mistral
Question. What city did Super Bowl 50 take place in?
Ground truth. *Santa Clara*
Baseline ($f = 0.942$): The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.
HCPC-v1 ($f = 0.478$): San Francisco Bay Area
HCPC-v2 ($f = 0.942$, recovered): The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

Case 3. *Dataset:* squad *Model:* mistral
Question. What year did the Denver Broncos secure a Super Bowl title for the third time?
Ground truth. *2015*
Baseline ($f = 0.795$): The provided context does not specify the exact year when the Denver Broncos secured a Super Bowl title for the third time. However, it is implied that this event happened in Super Bowl 50, which was held in 2016.
HCPC-v1 ($f = 0.485$): I cannot find this information in the provided context. The context only mentions that the Denver Broncos won the AFC Championship Game and secured home-field advantage throughout the AFC playoffs, but it does not speci\{}dots
HCPC-v2 ($f = 0.795$, recovered): The provided context does not specify the exact year when the Denver Broncos secured a Super Bowl title for the third time. However, it is implied that this event happened in Super Bowl 50, which was held in 2016.

Case 4. *Dataset:* squad *Model:* llama-3.3-70b
Question. If Roman numerals were used, what would Super Bowl 50 have been called?
Ground truth. *Super Bowl L*
Baseline ($f = 0.502$): If Roman numerals were used, Super Bowl 50 would have been called Super Bowl L.
HCPC-v1 ($f = 0.491$): Super Bowl L.
HCPC-v2 ($f = 0.502$, recovered): If Roman numerals were used, Super Bowl 50 would have been called Super Bowl L.

Figure 6: **Real-world case studies of the refinement paradox.** Five queries where the same question is answered correctly with raw retrieval, incorrectly after per-passage refinement, and correctly again under coherence-gated refinement. Faithfulness f is the NLI entailment score (higher is better; ≥ 0.5 is the conventional faithful threshold). All cases satisfy baseline = faithful, HCPC-v1 = hallucinated, HCPC-v2 = faithful.

mode) and `v2_recovery` = $\text{Faith}_{\text{v2}} - \text{Faith}_{\text{v1}}$ (the effectiveness of selective gating). Per-query NLI scores are written to `results/multidataset/summary.csv`; the aggregate table is generated deterministically from that file.

The paradox is considered to replicate at six-dataset scale if `paradox_drop` is positive on at least four of the six datasets, with the direction preserved on both the domain-matched (SQuAD, NaturalQs, TriviaQA) and domain-mismatched (PubMedQA, FinanceBench) subsets. HCPC-v2's

selective gating is considered to help if `v2_recovery` is positive on every dataset where `paradox_drop` is positive. We will report the outcome as written regardless of direction; a flat or inverted pattern on specific datasets is informative because it tells us where the coherence account stops generalizing.

J.3 Generator stratification (Item A3)

The same multi-dataset query set is rerun across three generators (`src/generators.py` registry):

- **Mistral-7B-Instruct-v0.2** (Mistral AI). Primary generator.
- **Llama-3-8B-Instruct** (Meta). Cross-architecture replication.
- **Qwen2.5-7B-Instruct** (Alibaba). Tests whether the paradox is specific to Western-pretrained model families.

The generator-stratified table (`results/multidataset/by_generator.csv`) reports the same three faithfulness conditions on the multidataset query set, separately for each generator. The paradox is called *generator-invariant* if the sign of `paradox_drop` is preserved across all three generators on at least four of the six datasets; a sign flip on any row localizes the paradox to a specific (dataset, generator) cell rather than to the method as such. We will report the full cell-level table with confidence intervals regardless of outcome, and call out any cell where the three generators disagree on sign.

J.4 Reproducibility checkpoints

The multi-dataset harness (`experiments/run_multidataset_validation.py`) is checkpointed at the (dataset, generator, condition) tuple level: each completed tuple is appended to `completed_tuples.json` with its per-query CSV, and the runner re-enters by skipping completed tuples. This makes it safe to re-run after partial failure (e.g., Self-RAG unavailable on M4) without recomputing finished tuples.

K Head-to-head: HCPC-v2 vs Self-RAG and CRAG

The previous draft of this paper avoided direct comparison against external retrieval-augmentation methods on the grounds that our contribution is the failure-mode analysis rather than a new state-of-the-art system. Reviewers correctly noted that this leaves the question of whether HCPC-v2's advantage survives against published selective-refinement systems unanswered. This section addresses that gap by running Self-RAG and CRAG on the same query set, with the same retrieval corpus and top-k, alongside our four conditions.

K.1 Conditions

Five conditions are evaluated end-to-end on the multidataset query set:

Baseline fixed 1024-token chunks, standard top-$k = 3$ retrieval, no refinement, Mistral-7B (Ollama) generator.

HCPC-v1 OR-gate selective refinement on the same retrieval. Always-refine when either similarity or cross-encoder gate triggers.

HCPC-v2 AND-gate + rank protection + merge-back, as described in §6.

CRAG our reimplementation of Yan et al. [2024]. The published evaluator is a private fine-tuned T5-large; we substitute a cross-encoder (`ms-marco-MiniLM-L-6-v2`) and threshold its logits to recover the three-bucket confidence signal (`correct` / `ambiguous` / `incorrect`). Knowledge refinement decomposes each retained passage into sentence-level strips and keeps the top-k scoring strips. Web-search augmentation is disabled by default (closed-corpus comparison); the substitution and the disabled web path are documented transparently in `src/crag_retriever.py`.

Self-RAG the published `selfrag/selfrag_llama2_7b` checkpoint [Asai et al., 2024], fed the same retrieval passages and prompted with the model's training-time format (`[Retrieve]<paragraph>...</paragraph>`). Reflection tokens (`[Relevant]` / `[Fully supported]` / `[Utility:N]`) are parsed and reported as diagnostics. Self-RAG inference requires a CUDA GPU; the harness runs on M4 for the four other conditions and is re-invoked on a GPU host to fill in the Self-RAG rows.

All conditions share: the same six datasets (§J), the same retrieval corpus, the same NLI-based faithfulness scorer, and identical top-$k = 3$.

K.2 Preregistered reporting plan

For each of the six datasets we will report two paired quantities per condition: the NLI-measured faithfulness rate and the binary hallucination rate. That is a $6 \times 2 \times 5$ grid of cells, populated from `results/headtohead/summary.csv`, with the best condition per (dataset, metric) marked in bold and 95% bootstrap confidence intervals reported alongside the point estimate. All five conditions share the same retrieval corpus, top-k, and NLI scorer so that differences are attributable to the refinement policy rather than the upstream pipeline. We commit to releasing per-query records so that any conditional metric a reader wants (e.g., faithfulness on the subset where Self-RAG emits `[Fully supported]`) can be recomputed without re-running inference.

K.3 Diagnostics

Beyond aggregate metrics, the harness logs condition-specific decision traces to `results/headtohead/diagnostics.`

- **CRAG:** per-passage labels and the resulting query-level decision (`correct` / `ambiguous` / `incorrect`), the number of strips retained, and the evaluator logits. These let us inspect whether CRAG's evaluator triggers on the same queries our coherence signals flag.
- **Self-RAG:** reflection-token counts per query and the parsed `[Relevant]` / `[Fully supported]` / `[Utility:N]` judgements. We report the conditional hallucination rate when Self-RAG self-judges its output as `Fully supported` versus when it does not — a measure of the calibration of the model's own faithfulness self-assessment.

K.4 What HCPC-v2 has to beat

We will treat the comparison as a success on three explicit conditions:

1. HCPC-v2's faithfulness is within 1 pp of Self-RAG's on the domain-matched datasets (SQuAD, NaturalQs, TriviaQA), *and* clearly better than CRAG-with-disabled-web.

2. HCPC-v2's coherence-paradox signal (faith_drop on HCPC-v1 relative to baseline) reproduces the same direction we report on the original 2-dataset study, on at least 4 of 6 datasets.
3. HCPC-v2's Self-RAG-style *self-support rate* (the fraction of queries on which Self-RAG itself emits `[Fully supported]`) is correlated positively with our CCS metric on matched queries, validating that CCS captures something the Self-RAG model also tracks internally.

Honest negative outcomes. If Self-RAG dominates HCPC-v2 on all six datasets, the paper's contribution narrows to the failure-mode diagnosis (which is independently valuable but does not justify HCPC-v2 as a deployment recommendation); we will state this outcome plainly in the discussion. If CRAG with web search disabled is competitive with Self-RAG, the practical implication is that retrieval refinement matters more than reflection-token training, which would also be worth reporting as written.